

E-Commerce Sales & Customer Analytics

Dataset

Use the **Brazilian E-Commerce Public Dataset by Olist** on Kaggle. It contains about **100,000 orders from 2016–2018**, with order, product, customer, payment, freight, seller and review information.

[Kaggle — Olist Brazilian E-Commerce Dataset](#)

Business problem

“Management wants to understand sales performance, customer behavior, product performance, and delivery experience to improve revenue and customer satisfaction.”

Tools

SQL | Python | Pandas | Excel | Power BI | DAX

Data preparation

Downloaded Tables:

- olist_orders_dataset.csv
- olist_order_items_dataset.csv
- olist_order_payments_dataset.csv
- olist_order_reviews_dataset.csv
- olist_products_dataset.csv
- olist_customers_dataset.csv
- olist_sellers_dataset.csv
- olist_product_category_name_translation.csv

Python tasks

- Inspect shape and data types

```
=====
CUSTOMERS
=====
Shape: (99441, 5)

Data Types:
customer_id          str
customer_unique_id   str
customer_zip_code_prefix  int64
customer_city        str
customer_state       str
dtype: object

First 5 rows:
   customer_id  ... customer_state
0  06b8999e2fba1fbc88172c00ba8bc7  ...  SP
1  18955e83d37fd6b2def6b18a428ac77  ...  SP
2  4e7b3e0b28858e6bd08712fdd0374a03  ...  SP
3  b2b6027bc5c5109e529d4dc6358b12c3  ...  SP
4  4f2d8ab171c80ec364f7c12e35b23ad  ...  SP

[5 rows x 5 columns]

=====
ORDERS
=====
Shape: (99441, 8)

Data Types:
order_id          str
customer_id       str
order_status      str
order_purchase_timestamp  str
order_approved_at  str
order_delivered_carrier_date  str
order_delivered_customer_date  str
order_estimated_delivery_date  str
dtype: object

First 5 rows:
   order_id  ... order_estimated_delivery_date
0  e481f51cddc54678b7cc49136f2d0aaf7  ...  2017-10-18 00:00:00
1  53c0b2fcb0c7dce0bb6741e2150273451  ...  2018-08-13 00:00:00
2  47770eb9108c20bc4494d0cf070c65d  ...  2018-09-04 00:00:00
3  949d5b44dbf5de918fe9c16f97b45f8a  ...  2017-12-15 00:00:00
4  ea21c59c0840e6cb83e9ceb5573f8159  ...  2018-02-26 00:00:00

=====
ORDER_ITEMS
=====
Shape: (112650, 7)

Data Types:
order_id          str
order_item_id     int64
product_id        str
seller_id         str
shipping_limit_date  str
price            float64
freight_value     float64
dtype: object

First 5 rows:
   order_id  order_item_id  ... price freight_value
0  00018242feb8c5a6d1ba2dd792cb16214  1  ...  58.98  13.29
1  00018f77f2f832bc557198d7a144bd3  1  ...  239.98  19.93
2  000229ec398224ef6ca0657da4fc703e  1  ...  199.00  17.87
3  00024acbcd0f0a6daa1e931b038114c75  1  ...  12.99  12.79
4  00042b2bcf59d7ce69dfabb4e55b4fd9  1  ...  199.98  18.14

=====
PAYMENTS
=====
Shape: (103886, 5)

Data Types:
order_id          str
payment_sequential  int64
payment_type       str
payment_installments  int64
payment_value      float64
dtype: object

First 5 rows:
   order_id  ... payment_value
0  b81ef226f3fe1789b1e8b2cac839d17  ...  99.33
1  a9810da82917af2d9aefd1278f1dcfa0  ...  24.39
2  25e9ea4e93396b6fa8d3dd780e76c1bd  ...  65.71
3  ba78997921bbcdcd1373bb41e913ab953  ...  187.78
4  42fdf880ba16b47b59251dd489d441a  ...  128.45

=====
REVIEWS
=====
Shape: (99224, 7)

Data Types:
review_id          str
order_id           str
review_score       int64
review_comment_title  str
review_comment_message  str
review_creation_date  str
review_answer_timestamp  str
dtype: object

First 5 rows:
   review_id  ... review_answer_timestamp
0  70c2406110b926393a56f80a40eba40  ...  2018-01-18 21:46:59
1  08e641a11e56f8c1ad469d5645fdfe  ...  2018-03-11 03:05:13
2  228ce5500dc1d8e020d8d1322874b6f0  ...  2018-02-18 14:36:24
3  e64fb393e7b32834bb789ff8bb30750e  ...  2017-04-21 22:02:06
4  f7c423c7fe1938f181bec41a392ddeb  ...  2018-03-02 10:26:53

=====
PRODUCTS
=====
Shape: (32951, 9)

Data Types:
product_id          str
product_category_name  str
product_name_lenght  float64
product_description_lenght  float64
product_photos_qty  float64
product_weight_g  float64
product_length_cm  float64
product_height_cm  float64
product_width_cm  float64
dtype: object

First 5 rows:
   product_id  ... product_width_cm
0  1e9e8ef04dbcff4541ed26657ae517e5  ...  14.0
1  3aa071139cb16b67ca9e5dea641aa2f  ...  20.0
2  96bd7aec8810374ed1b65e291975717f  ...  15.0
3  cef67b0cfe19066a932b7673e239eb23d  ...  26.0
4  9dc1a7de274444849c219cfff19508071  ...  13.0

=====
TRANSLATION
=====
Shape: (71, 2)

Data Types:
product_category_name  str
product_category_name_english  str
dtype: object

First 5 rows:
   product_category_name  product_category_name_english
0  beleza_saude  health_beauty
1  informatica_acessorios  computers_accessories
2  automotivo  auto
3  cama_mesa_banho  bed_bath_table
4  moveis_decoracao  furniture_decor

=====
SELLERS
=====
Shape: (3095, 4)

Data Types:
seller_id          str
seller_zip_code_prefix  int64
seller_city        str
seller_state       str
dtype: object

First 5 rows:
   seller_id  ... seller_state
0  3442f8959a84dea7ee197c632cb2df15  ...  SP
1  d1b65cf7debc3361ea80b5f14c68d2e2  ...  SP
2  ce3a9de9e960102d0677a81f500bb7b2d  ...  RJ
3  c9f3eea2e14555b6faeea3dd58c1b1c3  ...  SP
4  51a04a8a6bdc23deccc82d0b80742cf  ...  SP

[5 rows x 4 columns]

=====
GEOLOCATION
=====
Shape: (1000163, 5)

Data Types:
geolocation_zip_code_prefix  int64
geolocation_lat             float64
geolocation_lng             float64
geolocation_city            str
geolocation_state           str
dtype: object

First 5 rows:
   geolocation_zip_code_prefix  ... geolocation_state
0  1037  ...  SP
1  1046  ...  SP
2  1046  ...  SP
3  1041  ...  SP
4  1035  ...  SP
```

- Identify primary/foreign keys

```
Primary Key Validation
-----
orders → order_id: Valid PK? True (Unique: True, No Nulls: True)
products → product_id: Valid PK? True (Unique: True, No Nulls: True)
customers → customer_id: Valid PK? True (Unique: True, No Nulls: True)
sellers → seller_id: Valid PK? True (Unique: True, No Nulls: True)
reviews → review_id: Valid PK? False (Unique: False, No Nulls: True)

Foreign Key Validation
-----
order_items → orders: True
order_items → products: True
order_items → sellers: True
payments → orders: True
reviews → orders: True
```

- Check duplicate IDs

```

Duplicate Check
-----
orders: 0 duplicate IDs
customers: 0 duplicate IDs
products: 0 duplicate IDs
sellers: 0 duplicate IDs
reviews: 814 duplicate IDs
order_items: 0 duplicate (order_id + order_item_id)

```

- Check missing values

<pre> Missing Value Check ----- customers Series([], dtype: int64) orders order_delivered_customer_date 2965 order_delivered_carrier_date 1783 order_approved_at 160 dtype: int64 order_items Series([], dtype: int64) payments Series([], dtype: int64) reviews review_comment_title 87656 review_comment_message 58247 dtype: int64 </pre>	<pre> products product_category_name 610 product_name_lenght 610 product_description_lenght 610 product_photos_qty 610 product_weight_g 2 product_length_cm 2 product_height_cm 2 product_width_cm 2 dtype: int64 sellers Series([], dtype: int64) translation Series([], dtype: int64) geolocation Series([], dtype: int64) </pre>
--	--

- Convert timestamps

```

Timestamps Conversion
-----
orders → order_purchase_timestamp: Converted Successfully
data Type: datetime64[us]

orders → order_approved_at: Converted Successfully
data Type: datetime64[us]

orders → order_delivered_carrier_date: Converted Successfully
data Type: datetime64[us]

orders → order_delivered_customer_date: Converted Successfully
data Type: datetime64[us]

orders → order_estimated_delivery_date: Converted Successfully
data Type: datetime64[us]

order_items → shipping_limit_date: Converted Successfully
data Type: datetime64[us]

```

- Calculate delivery time VS Review Score

```

Delivery Time VS Review Score
-----
review_score    orders    average_delivery_days    average_delay_days    average_order_value
1.00000         9316             21.34             -3.34             193.41
1.50000           8             17.53             -5.34             204.70
2.00000         2916             16.66             -7.94             170.25
2.50000          30             18.67             -6.82             137.44
3.00000         7916             14.25             -10.08            151.07
3.33333           1             22.03             -13.09             42.11
3.50000          23             13.28             -13.22            149.63
4.00000        18868             12.31             -11.68            154.66
4.33333           1              3.77             -10.29             42.77
4.50000          53             12.80             -12.36            118.29
5.00000        56698             10.68             -12.68            156.32

```

- Calculate delivery delay VS Order Value

Delivery Delay VS Order Value

	orders	average_order_value	average_delay_days
order_value_group			
(9.589, 61.85]	24127	42.92	-10.71
(61.85, 105.28]	24173	81.67	-11.05
(105.28, 176.26]	24059	137.08	-11.33
(176.26, 13664.08]	24117	377.80	-11.62

- Translate product categories

Translation

	product_category_name	product_category_name_english
0	beleza_saude	health_beauty
1	informatica_acessorios	computers_accessories
2	automotivo	auto
3	cama_mesa_banho	bed_bath_table
4	moveis_decoracao	furniture_decor
..	...	...
66	flores	flowers
67	artes_e_artesanato	arts_and_craftmanship
68	fraldas_higiene	diapers_and_hygiene
69	fashion_roupa_infanto_juvenil	fashion_childrens_clothes
70	seguros_e_servicos	security_and_services

- Identify cancelled/unavailable orders

Order Status

order_status	
delivered	96478
shipped	1107
canceled	625
unavailable	609
invoiced	314
processing	301
created	5
approved	2

Name: count, dtype: int64

- Create an analytical order-level table

```
=====
ANALYTICAL ORDER-LEVEL TABLE
=====

Rows: 99441
Unique Orders: 99441
Duplicate Order IDs: 0

Columns:
- order_id
- customer_id
- order_status
- order_purchase_timestamp
- order_approved_at
- order_delivered_carrier_date
- order_delivered_customer_date
- order_estimated_delivery_date
- is_delivered
- is_cancelled
- is_unavailable
- delivery_days
- estimated_delivery_days
- delivery_delay_days

- customer_unique_id
- customer_zip_code_prefix
- customer_city
- customer_state
- order_revenue
- freight_cost
- item_count
- total_order_value
- payment_value
- payment_installments
- review_score
- unique_products
- product_categories

First 5 Rows:
      order_id  ... product_categories
0  e481f51cbdc54678b7cc49136f2d6af7  ...      housewares
1  53cdb2fc8bc7dce9b6741e2158273451  ...      perfumery
2  47770eb9100c2d0c44946d9cf07ec65d  ...          auto
3  949d5b44dbf5de918fe9c16f97b45f8a  ...      pet_shop
4  ad21c59c0840e6cb83a9ceb5573f8159  ...      stationery

Missing Values:
delivery_days                2965
order_delivered_customer_date 2965
delivery_delay_days          2965
order_delivered_carrier_date  1783
unique_products              775
freight_cost                 775
total_order_value            775
product_categories           775
order_revenue                775
item_count                   775
review_score                 768
order_approved_at            160
payment_value                 1
payment_installments          1
is_delivered                  0
dtype: int64

Delivery Status:
delivery_status

Early                88649
Delayed              7827
Not Delivered        2965
Name: count, dtype: int64
```

- Export cleaned data

```
=====
SUCCESS
=====

Analytical table exported to: Data/processed/analytical_order_level.csv
```

Derived columns:

```
delivery_days
estimated_delivery_days
delivery_delay_days
order_revenue
freight_cost
total_order_value
review_score
```

SQL practice

Loaded the cleaned tables into PostgreSQL.

- Monthly revenue

	month timestamp without time zone	total_orders bigint	monthly_revenue numeric
1	2016-09-01 00:00:00	1	134.97
2	2016-10-01 00:00:00	265	40325.11
3	2016-12-01 00:00:00	1	10.90
4	2017-01-01 00:00:00	750	111798.36
5	2017-02-01 00:00:00	1653	234223.40
6	2017-03-01 00:00:00	2546	359198.85
7	2017-04-01 00:00:00	2303	340669.68
8	2017-05-01 00:00:00	3546	489338.25
9	2017-06-01 00:00:00	3135	421923.37
10	2017-07-01 00:00:00	3872	481604.52
11	2017-08-01 00:00:00	4193	554699.70
12	2017-09-01 00:00:00	4150	607399.67
13	2017-10-01 00:00:00	4478	648247.65
14	2017-11-01 00:00:00	7289	987765.37
15	2017-12-01 00:00:00	5513	726033.19
16	2018-01-01 00:00:00	7069	924645.00
17	2018-02-01 00:00:00	6555	826437.13
18	2018-03-01 00:00:00	7003	953356.25
19	2018-04-01 00:00:00	6798	973534.09
20	2018-05-01 00:00:00	6749	977544.69
21	2018-06-01 00:00:00	6099	856077.86
22	2018-07-01 00:00:00	6159	867953.46
23	2018-08-01 00:00:00	6351	838576.64

- Top product categories

	product_category text	total_orders bigint	total_revenue numeric
1	health_beauty	8578	1225057.06
2	watches_gifts	5455	1160214.09
3	bed_bath_table	9075	1001983.69
4	sports_leisure	7463	947860.58
5	computers_accessori...	6480	883490.88
6	furniture_decor	6105	691573.27
7	housewares	5641	608019.81
8	cool_stuff	3493	602475.98
9	auto	3775	576341.05
10	toys	3754	466616.45

- Average order value

	month timestamp without time zone	total_orders bigint	total_revenue numeric	average_order_value numeric
1	2016-09-01 00:00:00	1	143.46	143.46
2	2016-10-01 00:00:00	265	46490.66	175.44
3	2016-12-01 00:00:00	1	19.62	19.62
4	2017-01-01 00:00:00	750	127482.37	169.98
5	2017-02-01 00:00:00	1653	271239.32	164.09
6	2017-03-01 00:00:00	2546	414330.95	162.74
7	2017-04-01 00:00:00	2303	390812.40	169.70
8	2017-05-01 00:00:00	3546	566851.40	159.86
9	2017-06-01 00:00:00	3135	490050.37	156.32
10	2017-07-01 00:00:00	3872	566299.08	146.25
11	2017-08-01 00:00:00	4193	645832.36	154.03
12	2017-09-01 00:00:00	4150	701077.49	168.93
13	2017-10-01 00:00:00	4478	751117.01	167.73
14	2017-11-01 00:00:00	7289	1153364.20	158.23
15	2017-12-01 00:00:00	5513	843078.29	152.93
16	2018-01-01 00:00:00	7069	1077887.46	152.48
17	2018-02-01 00:00:00	6555	966168.41	147.39
18	2018-03-01 00:00:00	7003	1120598.24	160.02
19	2018-04-01 00:00:00	6798	1132878.93	166.65
20	2018-05-01 00:00:00	6749	1128774.52	167.25
21	2018-06-01 00:00:00	6099	1011978.29	165.93
22	2018-07-01 00:00:00	6159	1027807.28	166.88
23	2018-08-01 00:00:00	6351	985491.64	155.17

- Late delivery analysis

	total_delivered_orders bigint	late_orders bigint	early_orders bigint	on_time_orders bigint	late_delivery_percentage numeric
1	99441	0	88649	0	0.00

- Customer revenue ranking

	customer_unique_id text	total_orders bigint	total_revenue numeric	revenue_rank bigint
1	0a0a92112bd4c708ca5fde585afaa8...	1	13664.08	1
2	da122df9eeddfedc1dc1f5349a1a690c	2	7571.63	2
3	763c8b1c9c68a0229c42c9fc6f662b...	1	7274.88	3
4	dc4802a71eae9be1dd28f5d788ceb5...	1	6929.31	4
5	459bef486812aa25204be022145caa...	1	6922.21	5
6	ff4159b92c40ebe40454e3e6a7c35e...	1	6726.66	6
7	4007669dec559734d6f53e029e3609...	1	6081.54	7
8	eebb5dda148d3893cdfaf5b5ca3040c...	1	4764.34	8
9	48e1ac109decbb87765a3eade6854...	1	4681.78	9
10	c8460e4251689ba205045f3ea17884...	4	4655.88	10
11	edde2314cc30e864a128ac95d6b2...	1	4513.32	11
12	a229eba70ec1c2abef51f04987deb7...	1	4445.50	12
13	edf81e1f3070b9dac83ec83dacdbb9...	1	4194.76	13
14	fa562ef24d41361e476e748681810e...	1	4175.26	14
15	ca27f3dac28fb1063fadd424c9d95fa	1	4163.51	15
16	5e713be0853d8986528d7869a0811...	1	4042.74	16
17	58483a1c055dfb600f57c5b8671745...	1	4034.44	17
18	011875f0176909c5cf0b14a9138bb6...	1	4016.91	18
19	f0767ae738c3d90e7b737d7b8b8bb4...	1	3979.55	19
20	bc5e25094a7d51b6aee35236572e6...	1	3826.80	20

- Repeat customers

	repeat_customers bigint
1	2801

- Complete Customer Segmentation

Showing rows: 1 to 1000						
	customer_unique_id text	total_orders bigint	total_revenue numeric	average_order_value numeric	customer_segment text	revenue_rank bigint
1	0a0a92112bd4c708ca5fde585afaa8...	1	13664.08	13664.08	One-Time Customer	1
2	da122df9eeddfdc1dc1f5349a1a690c	2	7571.63	3785.82	Repeat Customer	2
3	763c8b1c9c68a0229c42c9fc6f662b...	1	7274.88	7274.88	One-Time Customer	3
4	dc4802a71eae9be1dd28f5d788ceb5...	1	6929.31	6929.31	One-Time Customer	4
5	459bef486812aa25204be022145caa...	1	6922.21	6922.21	One-Time Customer	5
6	ff4159b92c40ebe40454e3e6a7c35e...	1	6726.66	6726.66	One-Time Customer	6
7	4007669dec559734d6f53e029e3609...	1	6081.54	6081.54	One-Time Customer	7
8	eebb5dda148d3893cdfaf5b5ca3040c...	1	4764.34	4764.34	One-Time Customer	8
9	48e1ac109decbb87765a3ead68540...	1	4681.78	4681.78	One-Time Customer	9
10	c8460e4251689ba205045f3ea17884...	4	4655.88	1163.97	Loyal Customer	10
11	edde2314c6c30e864a128ac95d6b2...	1	4513.32	4513.32	One-Time Customer	11
12	a229eba70ec1c2abef51f04987deb7...	1	4445.50	4445.50	One-Time Customer	12
13	edf81e1f3070b9dac83ec83dacdbb9...	1	4194.76	4194.76	One-Time Customer	13
14	fa562ef24d41361e476e748681810e...	1	4175.26	4175.26	One-Time Customer	14
15	ca27f3dac28fb1063fadd424c9d95fa	1	4163.51	4163.51	One-Time Customer	15
16	5e713be0853d8966528d7869a0811...	1	4042.74	4042.74	One-Time Customer	16
17	58483a1c055dfb600f57c5b8671745...	1	4034.44	4034.44	One-Time Customer	17
18	011875f0176909c5cf0b14a9138bb6...	1	4016.91	4016.91	One-Time Customer	18
19	f0767ae738c3d90e7b737d7b8b8bb4...	1	3979.55	3979.55	One-Time Customer	19
20	bc5e25094a7d51b6aee35236572e6...	1	3826.80	3826.80	One-Time Customer	20
21	3fec1a05177762f279472253113d56...	1	3792.59	3792.59	One-Time Customer	21
22	5d09b0d82126457e2a8ebfb9c9a1ffc4	1	3736.22	3736.22	One-Time Customer	22

Power BI dashboard

Created 3 pages.

Page 1 — Executive Overview

KPIs:

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value
- Average Review Score
- On-Time Delivery %

Visuals:



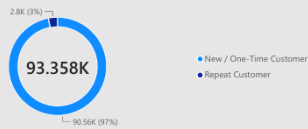
Page 2 — Customer Analytics

Visuals:

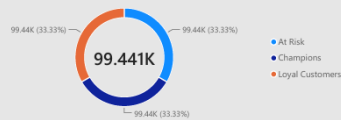
- One-Time vs repeat customers
- Revenue by customer segment
- Average order value
- Customer distribution by state
- Top 10 customers

Customer Analysis

One-Time vs Repeat Customers



Customer RFM Segmentation



Average Order Value
159.83

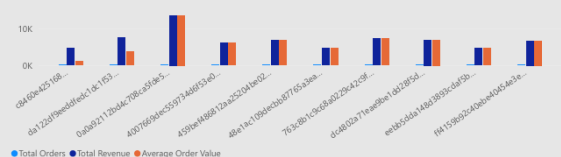
Revenue by Customer Segment



Customer Distribution by State



Top 10 Customers by Revenue



Added:

RFM segmentation

Recency
Frequency
Monetary

Created:

Champions
Loyal Customers
Potential Loyalists
At Risk
Lost Customers

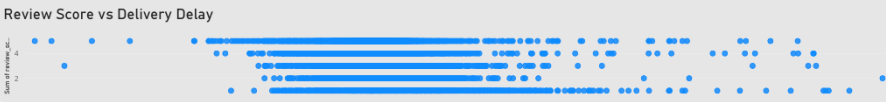
Page 3 — Delivery & Product Performance

Visuals:

- Average review score
- Review score vs delivery delay
- Delivery delay distribution
- Category-wise review score
- Top/bottom product categories

Delivery & Product Performance

Average Review Score
4.09



Delivery Delay Distribution



Category-wise Average Review Score



product_categories	Sum of Total Revenue	Sum of Total Orders	Average Review Score
	1,54,19,773.75	96478	3.16
agro_industry_and_commerce	1,54,19,773.75	96478	4.02
air_conditioning	1,54,19,773.75	96478	4.04
air_conditioning_computers_accessories	1,54,19,773.75	96478	3.00
art	1,54,19,773.75	96478	4.06
art_furniture_decor	1,54,19,773.75	96478	5.00
art_stationery	1,54,19,773.75	96478	1.00
arts_and_craftmanship	1,54,19,773.75	96478	4.29
arts_and_craftmanship_garden_tools	1,54,19,773.75	96478	1.00
audio	1,54,19,773.75	96478	3.85
audio_watches_gifts	1,54,19,773.75	96478	3.00
auto	1,54,19,773.75	96478	4.10
Total	1,54,19,773.75	96478	4.09

Top 10 Product Categories



Bottom 10 Product Categories

